

# Phase 4

## The Equivalent Circuit and Laboratory Tests

Electrical Machines II - Induction Machine Mastery

### Learning Target

After Phase 0 to Phase 3, the machine exists in your mind physically: stator RMF, rotor conductors, slip, rotor frequency, slots and harmonics. Phase 4 converts that physical machine into a **solvable per-phase electrical circuit**. The main target is:

Physical induction motor  $\rightarrow$  per-phase equivalent circuit  $\rightarrow$  test-based parameters

### 4.1 The Big Roadmap of Phase 4

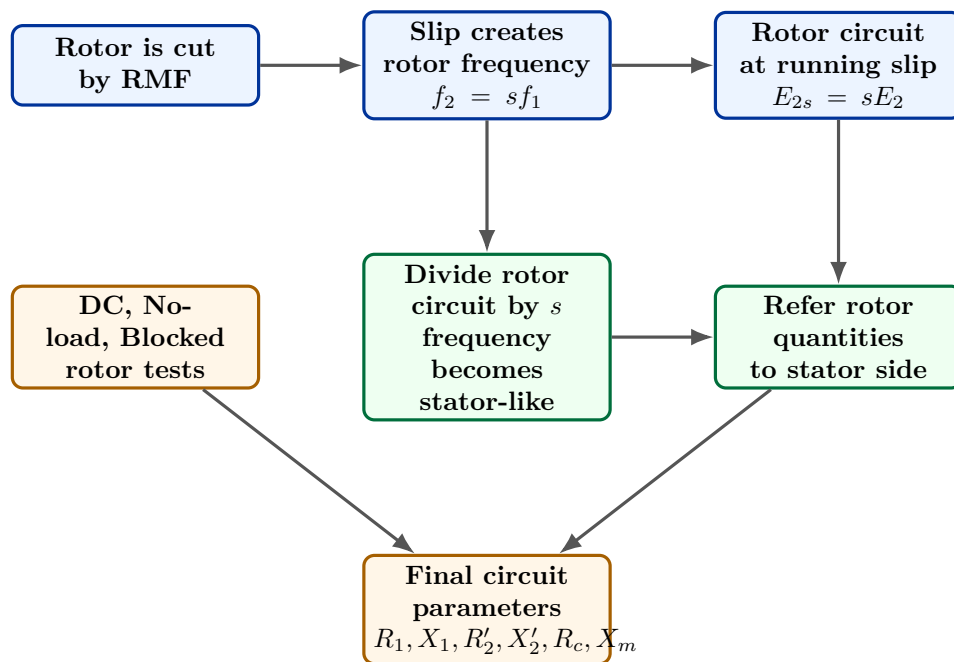


Figure 1: Phase 4 flow: from physical slip to circuit parameters. All stages are separated to avoid visual crowding.

### Exam Tip

In an exam, never start equivalent circuit derivation directly from the final circuit. Start from the running rotor facts:

$$E_{2s} = sE_2, \quad X_{2s} = sX_2, \quad f_2 = sf_1.$$

Then show the division by  $s$ . That is the heart of the derivation.

### 4.2 The Physical Meaning of Each Circuit Element

| Element              | Physical meaning                                                       | How it appears in test or calculation                                             |
|----------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| $R_1$                | Stator winding copper resistance per phase.                            | Obtained from DC resistance test, then used for stator copper loss $3I_1^2 R_1$ . |
| $X_1$                | Stator leakage reactance due to leakage flux not crossing the air gap. | Usually found from blocked rotor test after splitting $X_{eq}$ .                  |
| $R_c$                | Core-loss resistance representing hysteresis and eddy current loss.    | Obtained from no-load test after subtracting stator copper and mechanical losses. |
| $X_m$                | Magnetizing reactance required to set up air-gap flux.                 | Obtained from no-load test from the magnetizing component of current.             |
| $R'_2$               | Rotor copper resistance referred to stator side.                       | Found from blocked rotor test using $R'_2 = R_{eq} - R_1$ .                       |
| $X'_2$               | Rotor leakage reactance referred to stator side.                       | Often assumed $X_1 = X'_2 = X_{eq}/2$ if no separation is given.                  |
| $R'_2 \frac{1-s}{s}$ | Mechanical load represented as an equivalent electrical resistance.    | This part consumes the air-gap power converted into mechanical power.             |

Table 1: Circuit dictionary: each mathematical element has a physical source.

### 4.3 Rotor Circuit at Standstill and at Running Slip

At standstill, the rotor is not moving. Therefore the relative speed between RMF and rotor is maximum. The rotor frequency equals supply frequency:

$$s = 1, \quad f_2 = f_1, \quad E_{2s} = E_2, \quad X_{2s} = X_2$$

When the rotor runs, the relative speed between RMF and rotor decreases. Therefore the induced rotor emf and rotor leakage reactance both reduce in proportion to slip:

$$f_2 = sf_1, \quad E_{2s} = sE_2, \quad X_{2s} = sX_2$$

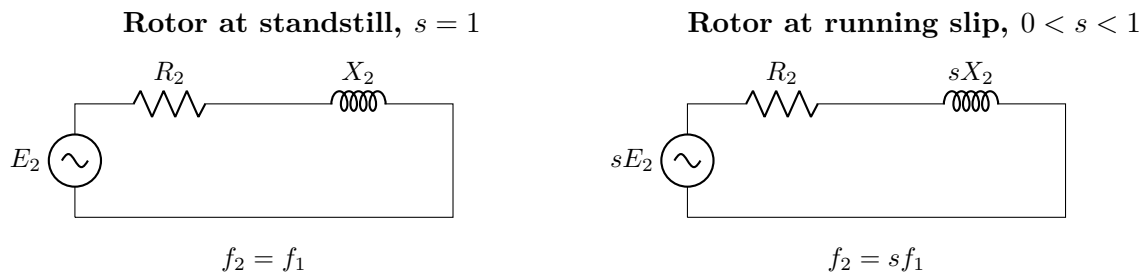


Figure 2: Rotor circuit before the mathematical frequency transformation. The two drawings are separated so labels do not overlap.

Rotor current at running slip is

$$I_2 = \frac{sE_2}{R_2 + jsX_2}. \quad (1)$$

This is a physically correct rotor-frequency circuit. But it is inconvenient because the rotor branch frequency changes with slip.

## 4.4 The Most Important Trick: Divide by Slip

Divide numerator and denominator of Eq. (1) by  $s$ :

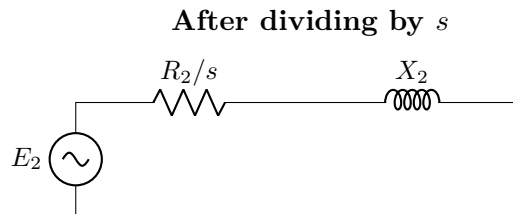
$$I_2 = \frac{sE_2}{R_2 + jsX_2} \cdot \frac{1/s}{1/s} \quad (1)$$

$$= \frac{E_2}{\frac{R_2}{s} + jX_2}. \quad (2)$$

### Meaning of Division by $s$

The division by  $s$  is not magic. It simply rewrites the running rotor circuit in a form that behaves like a stationary-frequency circuit. This allows the induction motor to be treated like a transformer with a special slip-dependent rotor resistance.

Rotor resistance becomes  $\frac{R_2}{s}$



Same current  $I_2$ , but the circuit is now expressed at supply-frequency form.

Figure 3: The transformed rotor circuit used for equivalent circuit development.

Now split the slip-dependent resistance:

$$\frac{R_2}{s} = R_2 + \left( \frac{R_2}{s} - R_2 \right) \quad (2)$$

$$= R_2 + R_2 \left( \frac{1-s}{s} \right). \quad (3)$$

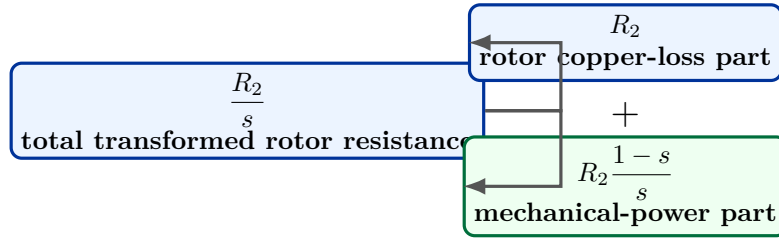


Figure 4: The transformed rotor resistance separates into copper-loss and mechanical-output parts.

#### Exam Tip

For a neat derivation answer, write this line very clearly:

$$\frac{R'_2}{s} = R'_2 + R'_2 \left( \frac{1-s}{s} \right)$$

The first term gives rotor copper loss. The second term represents the mechanical power developed.

## 4.5 Why Mechanical Load Becomes a Resistance

In the transformed rotor branch,

$$P_{ag} = 3I_2'^2 \frac{R'_2}{s}.$$

The rotor copper loss is

$$P_{rcl} = 3I_2'^2 R'_2.$$

Therefore the mechanical power developed is

$$P_{dev} = P_{ag} - P_{rcl} \quad (3)$$

$$= 3I_2'^2 \frac{R'_2}{s} - 3I_2'^2 R'_2 \quad (4)$$

$$= 3I_2'^2 R'_2 \left( \frac{1-s}{s} \right). \quad (4)$$

So the circuit element which represents converted mechanical power is

$$R_{mech} = R'_2 \left( \frac{1-s}{s} \right).$$

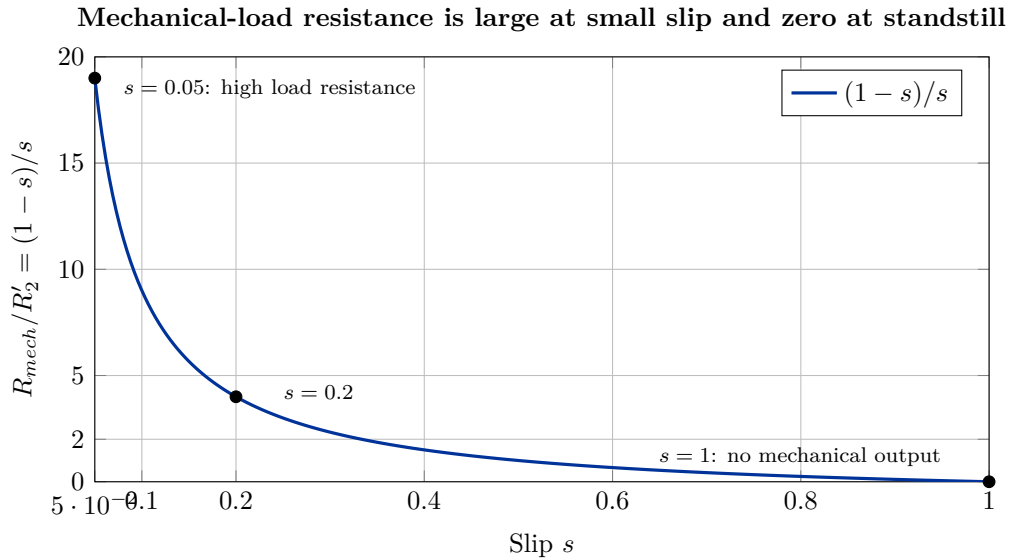


Figure 5: Graph of equivalent mechanical resistance against slip. Labels are kept outside the curve for readability.

### Common Mistake

Do not say that  $R'_2(1-s)/s$  is a real shaft resistor. It is not physically present. It is only a circuit representation of mechanical power conversion.

## 4.6 Referring Rotor Quantities to Stator Side

The induction motor is like a transformer with a rotating secondary. To combine stator and rotor circuits, rotor quantities are referred to the stator side. If the effective turns ratio is

$$a = \frac{N_1}{N_2},$$

then the referred quantities are

$$E'_2 = aE_2, \quad I'_2 = \frac{I_2}{a}, \quad R'_2 = a^2 R_2, \quad X'_2 = a^2 X_2.$$

After referring to the stator side, the rotor branch becomes

$$R'_2 + jX'_2 + R'_2 \left( \frac{1-s}{s} \right)$$

or equivalently

$$\frac{R'_2}{s} + jX'_2.$$

## 4.7 Complete Per-Phase Equivalent Circuit

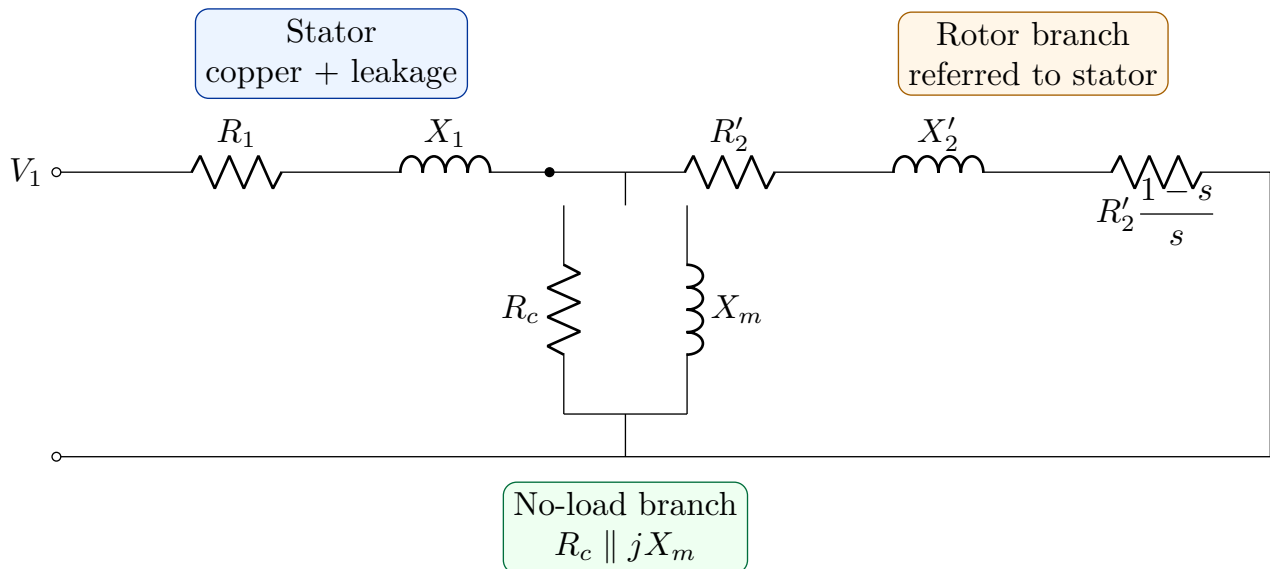


Figure 6: Exact per-phase equivalent circuit of a three-phase induction motor. The circuit is resized to text width to prevent margin overflow.

### Two Equivalent Rotor Forms

Both forms are accepted:

$$R'_2 + jX'_2 + R'_2 \left( \frac{1-s}{s} \right) \iff \frac{R'_2}{s} + jX'_2.$$

Use the split form when explaining power conversion. Use the compact form when solving current and torque problems.

## 4.8 Approximate Equivalent Circuit for Fast Numericals

For many exam numericals, the no-load branch is moved to the input side or neglected for approximate current calculations. This gives a simpler series path:

$$Z_{eq} = R_1 + \frac{R'_2}{s} + j(X_1 + X'_2).$$

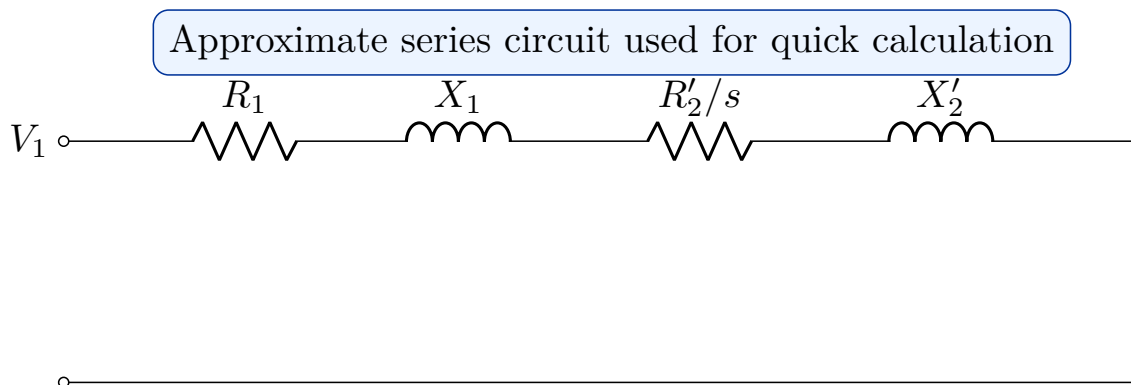


Figure 7: Approximate per-phase circuit. This is useful when magnetizing current is small compared with load current.

## 4.9 Laboratory Parameter Extraction: Overview

| Test                      | Machine condition                                           | Main measured quantities   | Parameters obtained          |
|---------------------------|-------------------------------------------------------------|----------------------------|------------------------------|
| <b>DC resistance test</b> | Rotor free, DC applied to stator terminals                  | $V_{dc}, I_{dc}$           | $R_1$ per phase              |
| <b>No-load test</b>       | Rated voltage and frequency, motor uncoupled                | $V_{0L}, I_0, P_0$         | $R_c, X_m$ , constant losses |
| <b>Blocked rotor test</b> | Rotor locked, low voltage applied until rated current flows | $V_{br,L}, I_{br}, P_{br}$ | $R'_2, X_1 + X'_2$           |

Table 2: Three basic tests and the parameters extracted from them.

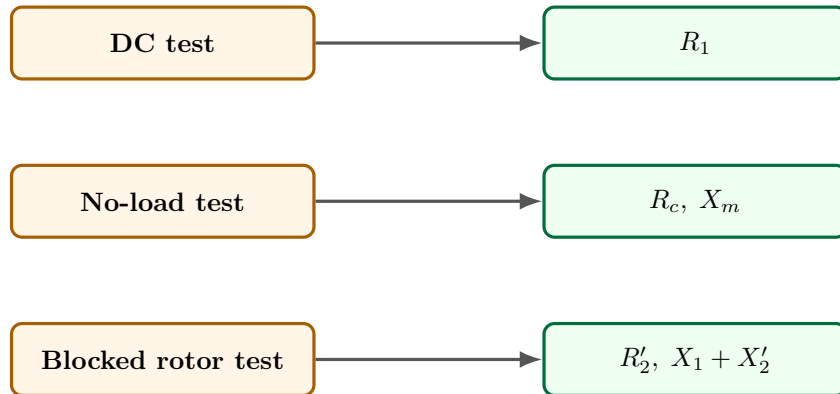
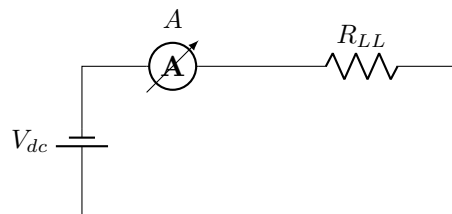


Figure 8: Parameter extraction flowchart. The outputs are placed far from the test boxes to avoid overlap.

## 4.10 DC Resistance Test

**Purpose:** Determine the stator resistance per phase  $R_1$ .



DC supply is applied between two stator line terminals

Figure 9: DC resistance test arrangement. The measured line-to-line resistance is  $R_{LL} = V_{dc}/I_{dc}$ .

Measured resistance between two line terminals is

$$R_{LL} = \frac{V_{dc}}{I_{dc}}.$$

For a star-connected stator:

$$R_1 = \frac{R_{LL}}{2}.$$

For a delta-connected stator:

$$R_1 = \frac{3R_{LL}}{2}.$$

### Common Mistake

Do not put the measured line-to-line resistance directly as  $R_1$ . Convert it to per-phase stator resistance according to the winding connection.

## 4.11 No-Load Test

**Purpose:** Determine the shunt branch parameters  $R_c$  and  $X_m$ , and estimate constant losses.

**Condition:** Rated voltage and rated frequency are applied. The motor shaft is unloaded, so slip is very small:

$$s \approx 0.$$

At very small slip,

$$R'_2 \frac{1-s}{s} \rightarrow \text{very large}.$$

So the rotor branch takes very little current and the input current is mainly exciting current.

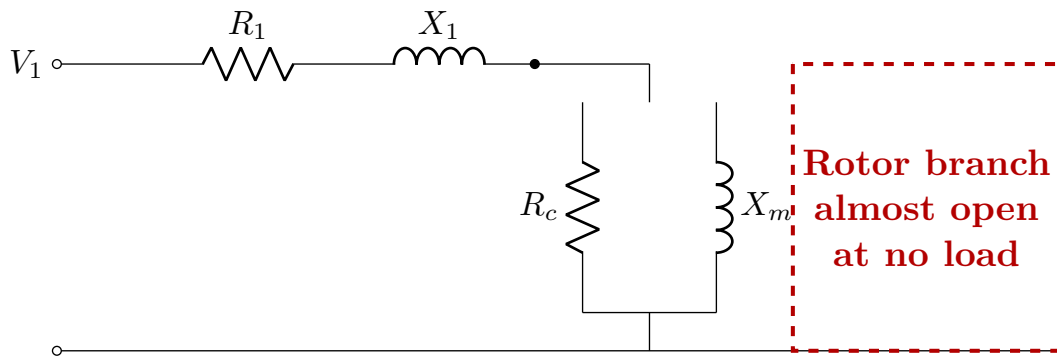


Figure 10: No-load equivalent circuit: magnetizing branch dominates because slip is nearly zero.

Let no-load readings be  $V_{0L}$ ,  $I_0$ , and total three-phase input power  $P_0$ .

For star connection:

$$V_{0ph} = \frac{V_{0L}}{\sqrt{3}}, \quad I_{0ph} = I_0.$$

For delta connection:

$$V_{0ph} = V_{0L}, \quad I_{0ph} = \frac{I_0}{\sqrt{3}}.$$



Stator copper loss at no load:

$$P_{sc0} = 3I_{0ph}^2 R_1.$$

Power left for core loss plus mechanical losses:

$$P_{const} = P_0 - P_{sc0}.$$

If friction and windage loss  $P_{fw}$  is known, then

$$P_c = P_{const} - P_{fw}.$$

Core-loss current per phase:

$$I_w = \frac{P_c}{3V_{0ph}}.$$

Magnetizing current:

$$I_m = \sqrt{I_{0ph}^2 - I_w^2}.$$

Therefore,

$$R_c = \frac{V_{0ph}}{I_w}, \quad X_m = \frac{V_{0ph}}{I_m}.$$

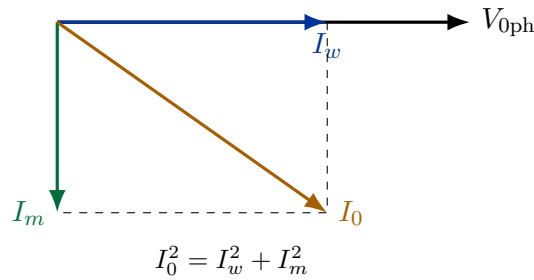


Figure 11: No-load current resolution into core-loss and magnetizing components. The labels are placed away from the vector lines.

## 4.12 Blocked Rotor Test

**Purpose:** Determine series parameters  $R'_2$  and  $X_{eq} = X_1 + X'_2$ .

**Condition:** Rotor is mechanically locked. Therefore

$$N_r = 0, \quad s = 1.$$

A low voltage is applied until rated current flows. Since the voltage is low, the shunt branch current is small and is usually neglected.

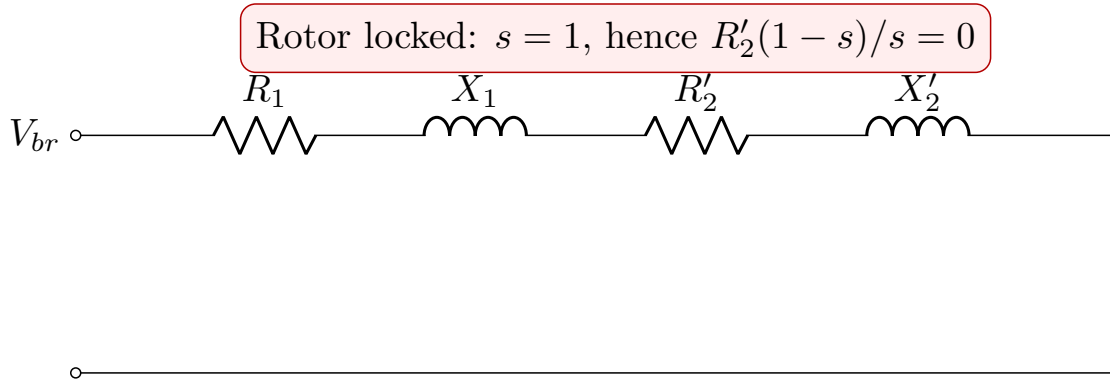


Figure 12: Blocked rotor equivalent circuit. The mechanical-load resistance is zero at  $s = 1$ .

Let blocked rotor readings be  $V_{br,L}$ ,  $I_{br}$ , and total three-phase power  $P_{br}$ .

Convert to per-phase values:

$$V_{br,ph} = \begin{cases} \frac{V_{br,L}}{\sqrt{3}}, & \text{star connection,} \\ V_{br,L}, & \text{delta connection,} \end{cases} \quad I_{br,ph} = \begin{cases} I_{br}, & \text{star connection,} \\ \frac{I_{br}}{\sqrt{3}}, & \text{delta connection.} \end{cases}$$

Blocked rotor impedance per phase:

$$Z_{br} = \frac{V_{br,ph}}{I_{br,ph}}.$$

Equivalent resistance per phase:

$$R_{eq} = \frac{P_{br}}{3I_{br,ph}^2}.$$

Equivalent reactance per phase:

$$X_{eq} = \sqrt{Z_{br}^2 - R_{eq}^2}.$$

Therefore,

$$R'_2 = R_{eq} - R_1.$$

If leakage reactance is equally divided,

$$X_1 = X'_2 = \frac{X_{eq}}{2}.$$

### Common Mistake

In blocked rotor test, do not apply full rated voltage. The rotor is locked, so a small applied voltage can produce rated current.

## 4.13 Complete Numerical Template

### Template for Solving Parameter-Extraction Problems

1. Convert every line quantity into per-phase quantity depending on star or delta connection.
2. From DC test, calculate  $R_1$ .
3. From no-load test, calculate  $P_{sc0} = 3I_0^2 R_1$  and then separate  $R_c$  and  $X_m$ .
4. From blocked rotor test, calculate

$$Z_{br} = \frac{V_{br,ph}}{I_{br,ph}}, \quad R_{eq} = \frac{P_{br}}{3I_{br,ph}^2}, \quad X_{eq} = \sqrt{Z_{br}^2 - R_{eq}^2}.$$

5. Obtain

$$R'_2 = R_{eq} - R_1, \quad X_1 + X'_2 = X_{eq}.$$

6. Draw the final equivalent circuit before writing the final answer.

### Exam Tip

The most exam-friendly answer has three layers: **diagram first, formula second, substitution third**. Even if numerical data is long, the examiner can follow your logic.

## 4.14 Mini Worked Example

A three-phase star-connected induction motor gives the following test results:

$$V_{dc} = 20 \text{ V}, \quad I_{dc} = 10 \text{ A};$$

$$V_{br,L} = 90 \text{ V}, \quad I_{br} = 15 \text{ A}, \quad P_{br} = 900 \text{ W}.$$

Find  $R_1$ ,  $R'_2$ , and  $X_1 + X'_2$ .

**Step 1: DC test.**

$$R_{LL} = \frac{20}{10} = 2 \Omega.$$

For star connection,

$$R_1 = \frac{R_{LL}}{2} = 1 \Omega.$$

**Step 2: Blocked rotor per-phase quantities.**

$$V_{br,ph} = \frac{90}{\sqrt{3}} = 51.96 \text{ V}, \quad I_{br,ph} = 15 \text{ A}.$$

$$Z_{br} = \frac{51.96}{15} = 3.464 \Omega.$$

$$R_{eq} = \frac{900}{3(15)^2} = 1.333 \Omega.$$

$$X_{eq} = \sqrt{(3.464)^2 - (1.333)^2} = 3.197 \Omega.$$

Therefore,

$$R'_2 = R_{eq} - R_1 = 1.333 - 1 = 0.333 \Omega,$$

$$X_1 + X'_2 = 3.197 \Omega.$$

If equal leakage reactance is assumed,

$$X_1 = X'_2 = 1.598 \Omega.$$

## 4.15 Final Exam-Ready Summary

### One-Page Revision Formula Bank

$$s = \frac{N_s - N_r}{N_s}, \quad f_2 = sf_1, \quad E_{2s} = sE_2, \quad X_{2s} = sX_2.$$

$$I_2 = \frac{sE_2}{R_2 + jsX_2} = \frac{E_2}{R_2/s + jX_2}.$$

$$\frac{R'_2}{s} = R'_2 + R'_2 \left( \frac{1-s}{s} \right).$$

$$R_{mech} = R'_2 \left( \frac{1-s}{s} \right).$$

$$P_{ag} = 3I_2'^2 \frac{R'_2}{s}, \quad P_{rcl} = 3I_2'^2 R'_2, \quad P_{dev} = 3I_2'^2 R'_2 \left( \frac{1-s}{s} \right).$$

$$Z_{br} = \frac{V_{br,ph}}{I_{br,ph}}, \quad R_{eq} = \frac{P_{br}}{3I_{br,ph}^2}, \quad X_{eq} = \sqrt{Z_{br}^2 - R_{eq}^2}.$$

### Phase 4 Complete

Next Phase: Power, Torque and Performance Dynamics